

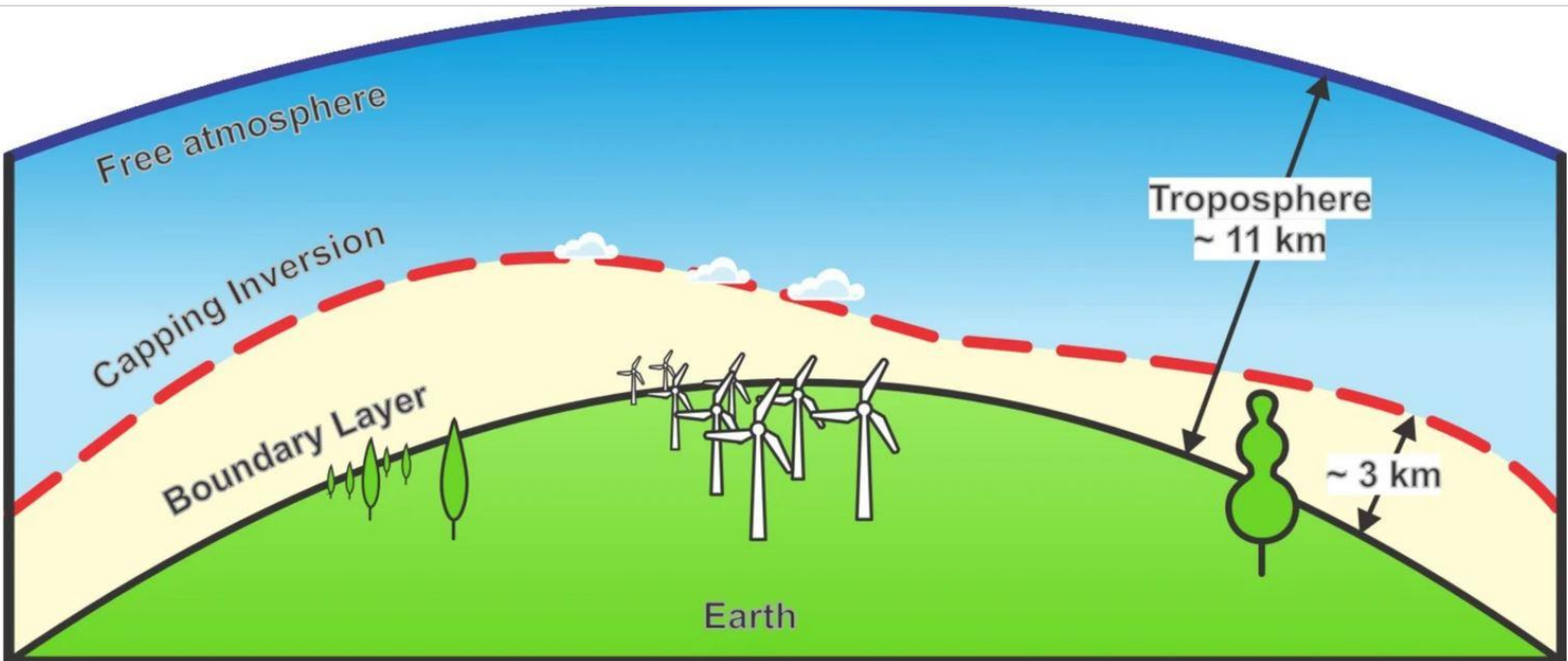
Planetary Boundary Layer Height Detection Using Ceilometer Data

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MOTIVATION

The Planetary Boundary Layer (PBL) is the lowest portion of Earth’s atmosphere, where surface heating and moisture fluxes drive vigorous mixing. Because this mixing governs particle dispersion, near-surface temperature gradients, and the triggering of convective storms, high-resolution observations of PBL height (PBLH) are critical for air-quality modeling, urban heat-island studies, and severe-weather nowcasting.

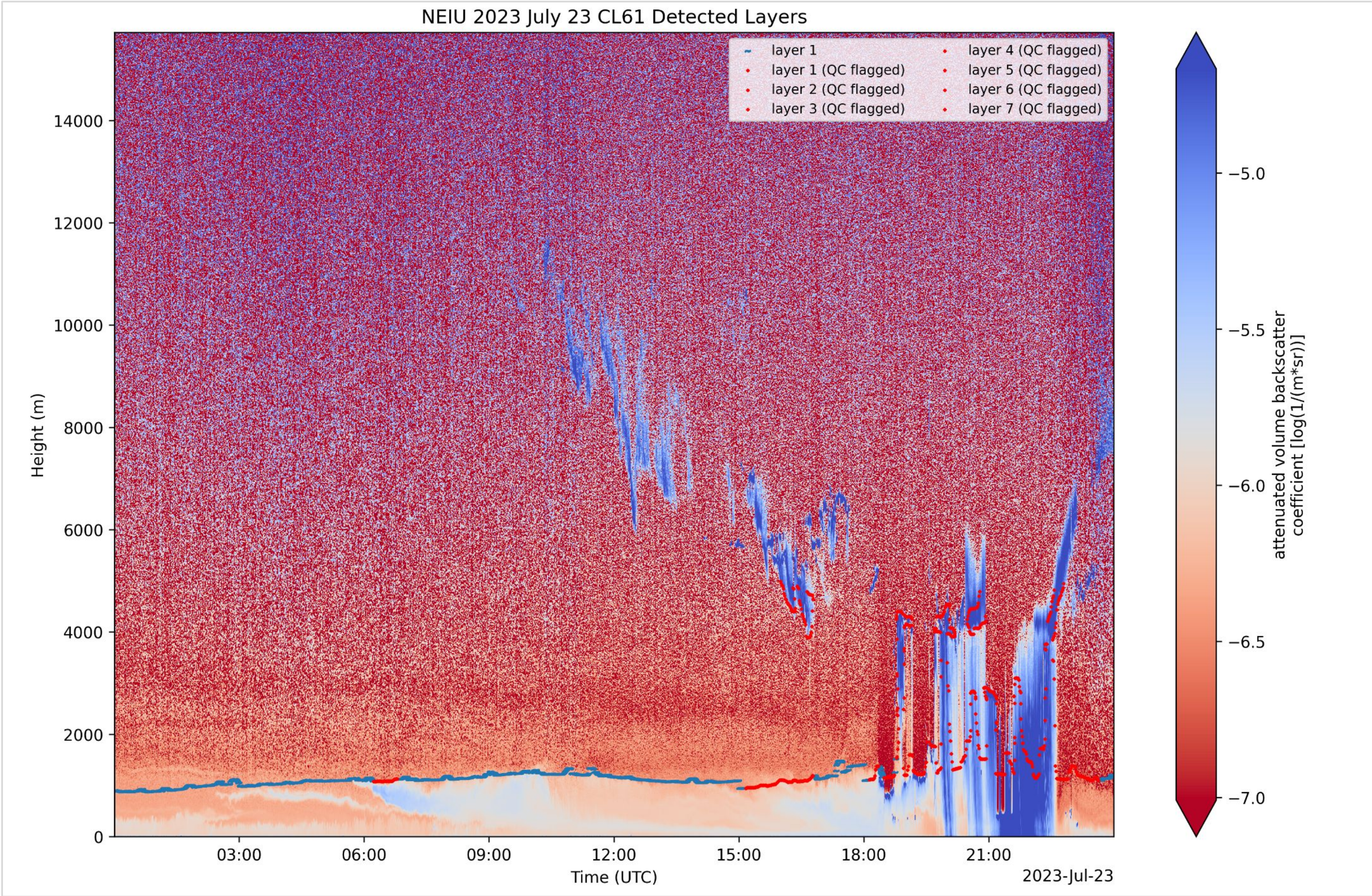
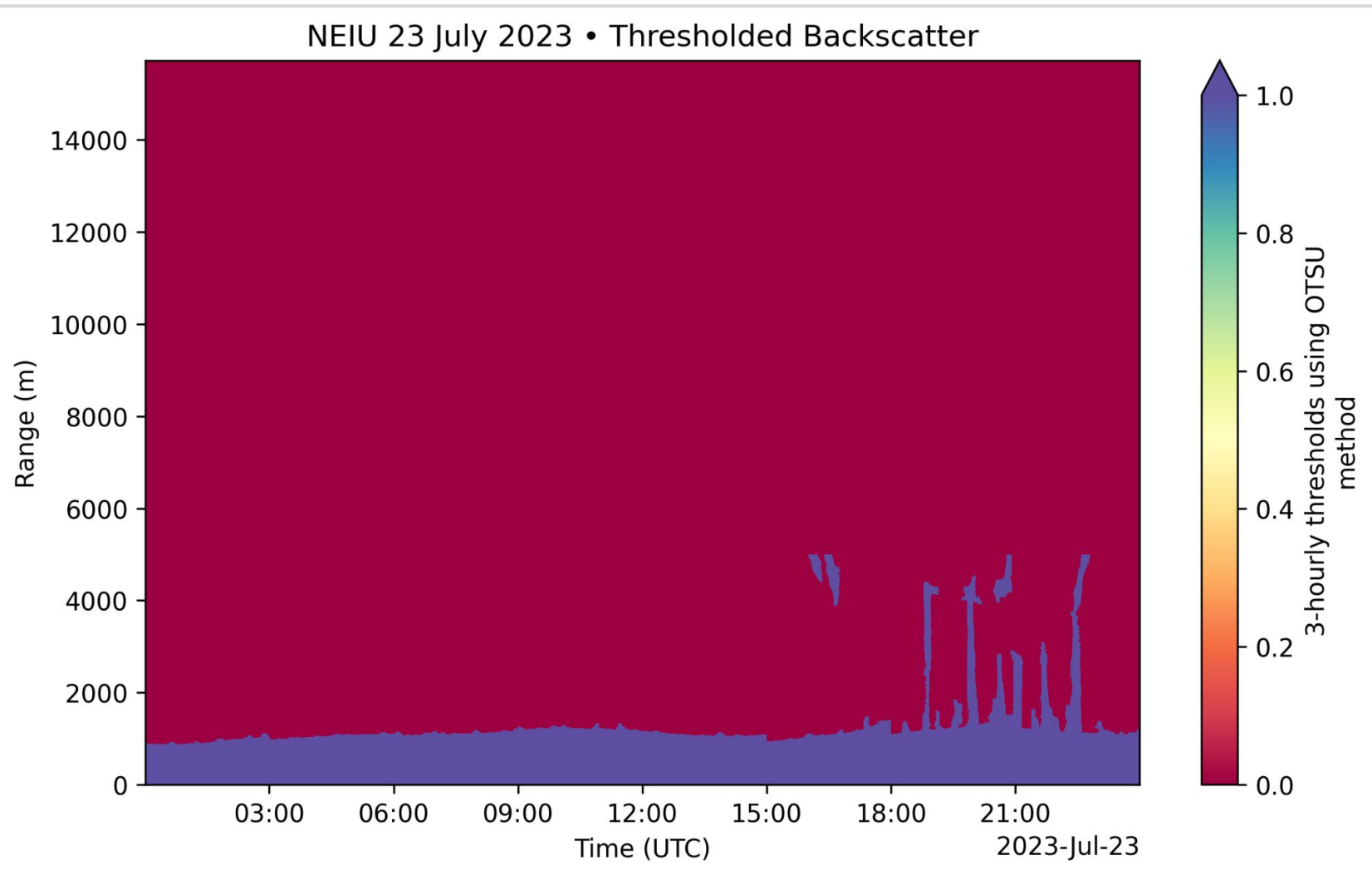
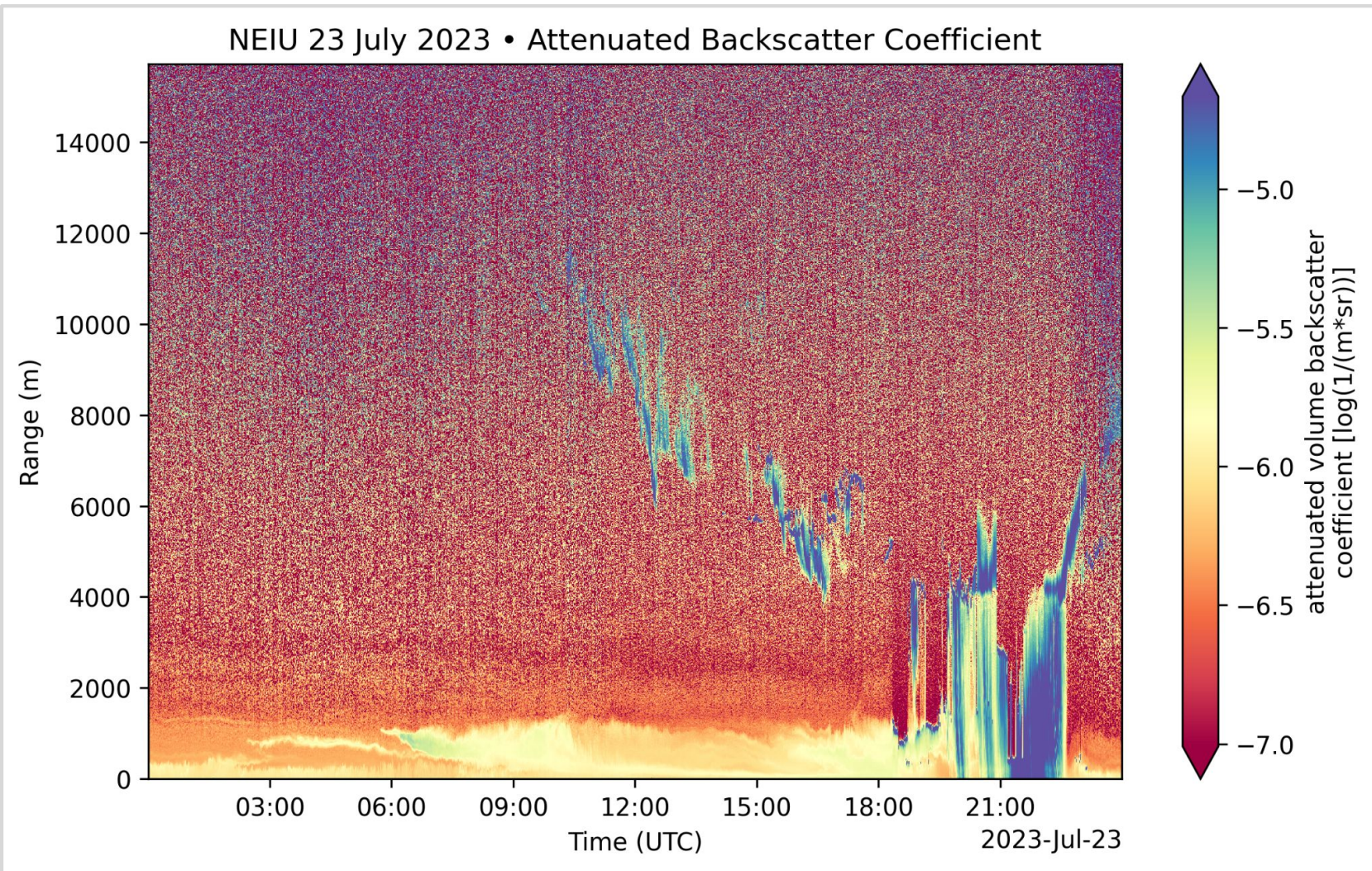
- Radiosondes, launched twice daily by the National Weather Service, remain the gold standard for detecting PBLH, but with a limited quantity of daily observations, they provide coarse data spatially and temporally. Ceilometers, a vertically oriented LIDAR device, provide backscatter measurements which can be used for PBLH detection. Automated Surface Observing Systems (ASOS) stations, located at many airports around the United States contain CL31 ceilometers, allowing for a much denser grid of measurements.



Vertical Representation of the Earth’s Surface, Boundary Layer, and Troposphere (Wallace and Hobbs 2006).

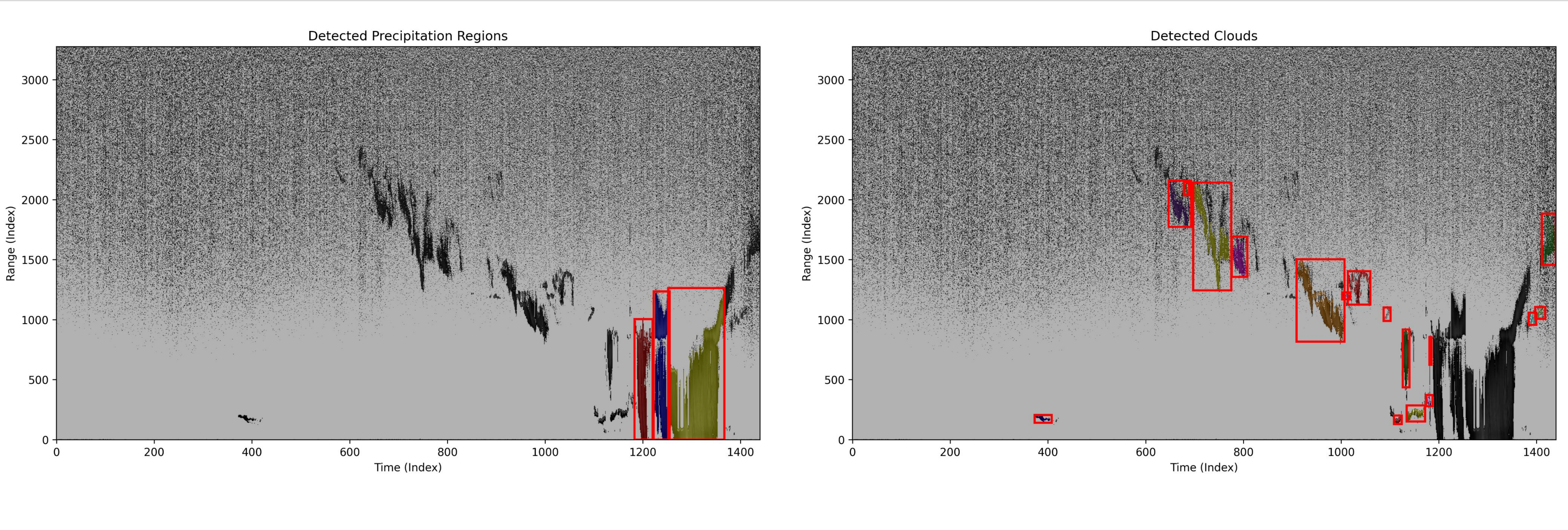
CHALLENGES

While the PBL is defined by turbulence, standard ceilometers only measure aerosol backscatter. Therefore, many features are subtle and challenging to differentiate. Additionally, attenuation with height causes an abundance of high frequency clutter. This is why a more complex algorithm is implemented, rather than simple gradient or edge detection.



METHODS

To allow for applications on any vertical LIDAR instrument, only backscatter is utilized. For noise removal, both a horizontal and vertical stationary (à trous) wavelet transform is applied with eight scales. From this, scales which ignore clutter are combined. Finally, automatic OTSU thresholding is applied for all points below 5000m, noisy regions are removed, and heights are extracted.



DELIVERABLES

1. Created an open source Python package
DanielWefer/ceil-feature-detection
2. PBLH dataset for ceilometer at NEIU, CCICS, and ATMOS

FUTURE DIRECTIONS

- Radiosonde and doppler LIDAR validation dataset
- Algorithm with stronger sensitivity towards the nocturnal PBL

EXPERIENCED GAINED

- Meteorological Instrumentation
- Computer Vision Techniques

QC HANDLING

High intensity features (fog, precipitation, and clouds) impact the automated PBLH thresholding. Manual thresholding and image segmentation are used to separate and label these regions. This method also allows for cloud base height to be calculated.

ADVANTAGES

- Applicable for many vertical range/time instruments (doppler lidar, HSRL, etc.)
- Robust noise handling

LIMITATIONS

- Biased towards the nocturnal residual layer
- Computationally slow

Wallace, J. M.; Hobbs, P. V. *Atmospheric Science: An Introductory Survey*, 2nd ed.; Academic Press: London, 2006. Muradyan P ; Anderson G ; Raut B ; O'Brien J ; Grover M ; Jackson R ; Tuftedal M ; Collis S ; Wawrzyniak E ; Sankaran R ; Shahkarami S ; Conrad N (2025): CROCUS Dual Polarization Ceilometer Data at Northeastern Illinois University Rooftop. Community Research on Climate and Urban Science Urban Integrated Field Laboratory (CROCUS UIFL), ESS-DIVE repository. Dataset. doi:10.15485/2510474 accessed via <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/2510474> on 2025-07-17